

Pratik Padole

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Professional Summary

Artificial Intelligence & Data Science undergraduate with strong foundations in Python, Java, Data Structures & Algorithms, Backend Development, and Machine Learning. Experienced in building AI-powered applications using FastAPI, LangChain, FAISS, and REST APIs. Passionate about developing scalable intelligent software systems and solving real-world engineering problems.

Technical Skills

Languages: Python, Java, JavaScript, SQL

AI & ML: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLMs, RAG

Frameworks/Libraries: FastAPI, Flask, LangChain, TensorFlow, Scikit-Learn, OpenCV, Pandas, NumPy, Matplotlib

Databases: MongoDB, MySQL, FAISS (Vector Database)

Tools: Git, GitHub, Postman, Docker

Education

Dr. D. Y. Patil College of Engineering, Akurdi, Pune

2023 – 2027

Bachelor of Engineering in Artificial Intelligence and Data Science

CGPA: 9.33 / 10

Projects

TalentMatch AI

Python, FastAPI, Sentence Transformers, Scikit-Learn, PyPDF

- Built an AI-powered Applicant Tracking System that ranks resumes against job descriptions using transformer-based semantic embeddings and cosine similarity.
- Developed an end-to-end NLP pipeline for PDF text extraction, preprocessing, skill extraction, batch embedding generation, and weighted candidate scoring.
- Designed RESTful APIs using FastAPI for multi-resume uploads, job description processing, and explainable candidate ranking.
- Optimized inference by caching job description embeddings, processing resume embeddings in batches, and containerized the application using Docker.

DocuMind AI

Python, FastAPI, LangChain, FAISS, Sentence Transformers, Gemini API, Docker

- Developed a Retrieval-Augmented Generation (RAG) system for intelligent document question answering using LangChain, FAISS, and Gemini API.
- Built an end-to-end pipeline for PDF text extraction, document chunking, embedding generation, semantic retrieval, and context-aware response generation.
- Designed RESTful APIs using FastAPI to support document upload, indexing, and natural language querying.
- Containerized the complete application using Docker, enabling scalable deployment and efficient document retrieval.

Student Performance Prediction System

Python, Scikit-Learn, Pandas, NumPy, Matplotlib

- Developed a machine learning system to predict student examination scores using academic and behavioral features.
- Performed data preprocessing, feature engineering, and exploratory data analysis.
- Trained and evaluated regression models including Linear Regression and Random Forest using MAE, MSE, and R^2 metrics.
- Built a modular prediction pipeline for real-time score estimation.

Relevant Coursework

Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, Machine Learning, Artificial Intelligence, Computer Vision

Achievements

- Solved 500+ Data Structures and Algorithms problems on LeetCode.
- Maintained a CGPA of 9.33/10 in Artificial Intelligence and Data Science.

Positions of Responsibility

Technical Lead – Society For Data Science (S4DS)

Jul 2025 – May 2026

- Conduct technical sessions and mentor students in backend development and project building.

Web Lead – Intel AI Student Club (IASC)

Aug 2025 – May 2026

- Lead web development initiatives for technical events, workshops, and student activities.